

Lab 1: Requirements Engineering & UML Use-Case Modelling

Self-Service Coffee Kiosk

Complete Solution

1. Requirements Table

The assignment requires exactly five Functional Requirements and two Nonfunctional Requirements, with measurable acceptance criteria and rationales.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a customer to select a coffee type.	High	When a customer selects Espresso, Americano, Latte, or Cappuccino, the chosen coffee is highlighted and retained in the order.	Core functionality required to start a coffee order.
FR-002	Functional	The system shall allow the customer to select a drink size.	High	When Small, Medium, or Large is selected, the chosen size is highlighted and the order price is updated accordingly.	Size is required for price calculation and preparation.
FR-003	Functional	The system shall allow customers to customize the selected coffee with available add-ons.	Medium	Selected add-ons such as extra shot, plant milk, or syrup appear in the order summary and affect the total price correctly.	Supports the customization options specified in the scenario.
FR-004	Functional	The system shall allow customers to pay for the order using a credit/debit card or mobile wallet.	High	A successful payment changes the order status to 'Paid' and records the selected payment method.	Payment is required before the kiosk can finalize the order.
FR-005	Functional	The system shall print a receipt containing order details and updated loyalty points after successful payment.	High	After successful payment, a receipt is printed containing the coffee, size, add-ons, amount paid, and loyalty points.	Provides order confirmation and the loyalty feature specified.
NFR-001	Nonfunctional	The system shall complete a single order from selection through payment in under 60 seconds.	High	At least 95% of valid test orders, measured from first selection to successful payment, complete in less than 60 seconds.	Ensures fast service during busy café periods.
NFR-002	Nonfunctional	The kiosk interface shall be straightforward and responsive during normal operation.	Medium	At least 95% of user interactions produce visible feedback within 2 seconds under normal operating load.	Reduces confusion and keeps the self-service experience efficient.

2. Actors and Use Cases

Actors

- Customer
- Payment Gateway
- Café Admin / Staff

Use Cases

ID	Use Case	Purpose
UC-01	Select Coffee	Customer chooses Espresso, Americano, Latte, or Cappuccino.
UC-02	Choose Size	Customer chooses Small, Medium, or Large.
UC-03	Customize Add-ons	Customer optionally adds extra shot, plant milk, or syrup.

ID	Use Case	Purpose
UC-04	Make Payment	Customer pays using card or mobile wallet.
UC-05	Print Receipt	System prints order details and loyalty points after payment.
UC-06	Update Menu / Prices	Admin updates coffee prices and menu information.
UC-07	Refill Ingredients	Admin records/refills available ingredients.
UC-08	View Sales Reports	Admin views sales reports.

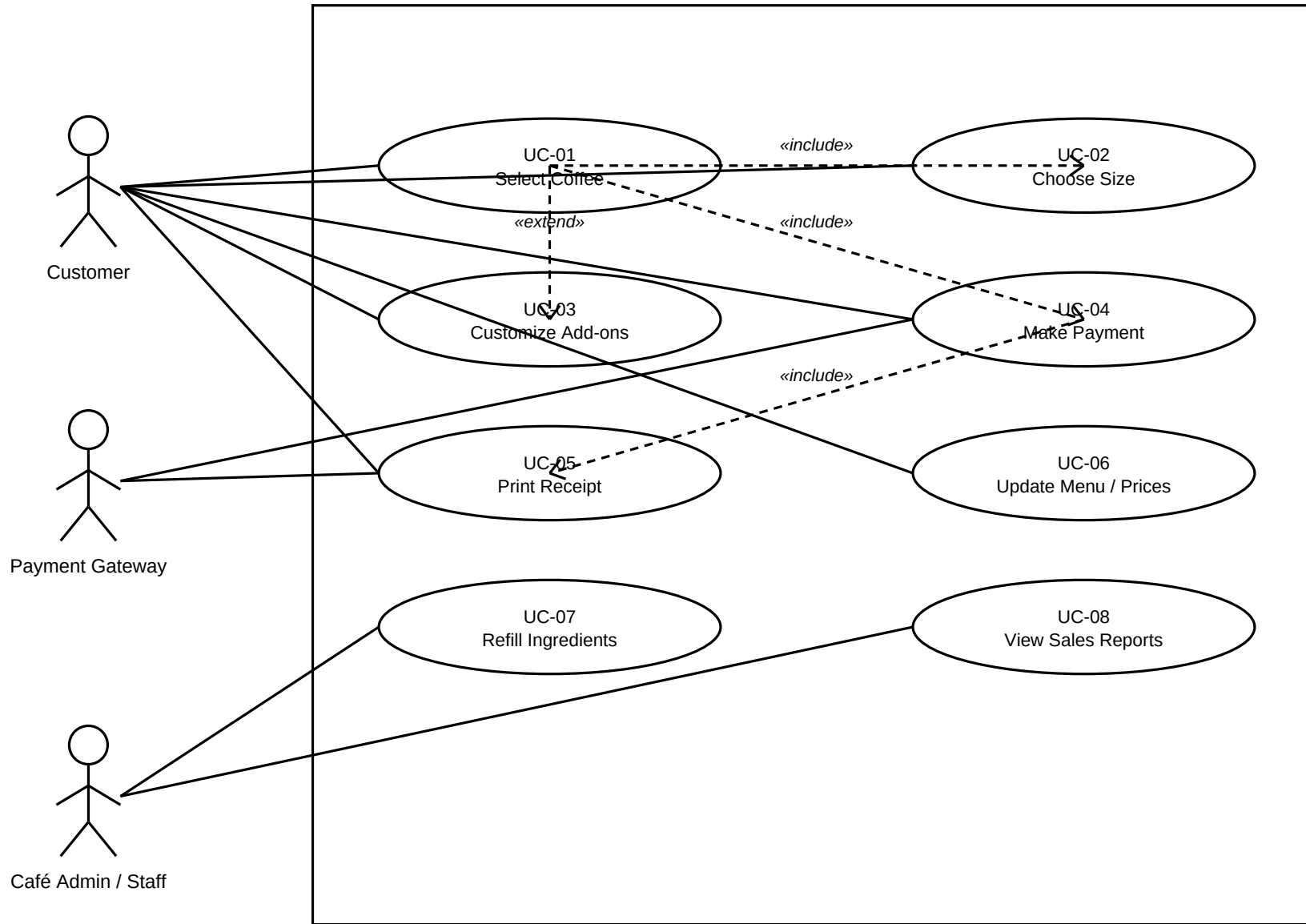
3. UML Use-Case Diagram

The diagram shows the system boundary, actors, use cases, associations, and a conditional «extend» relationship.

Diagram is provided on the following page.

UML Use-Case Diagram – Self-Service Coffee Kiosk

Self-Service Coffee Kiosk System



4. Use-Case Flow: Place Coffee Order

Main Success Scenario

1. Customer selects a coffee type from Espresso, Americano, Latte, or Cappuccino.
2. System displays the available sizes and the customer selects Small, Medium, or Large.
3. System displays available add-ons; the customer optionally selects extra shot, plant milk, or syrup.
4. System shows the complete order summary and total price.
5. Customer selects credit/debit card or mobile wallet as the payment method.
6. System sends the payment request to the Payment Gateway.
7. Payment Gateway authorizes the payment and returns the result.
8. System records the order as paid and calculates the applicable loyalty points.
9. System prints a receipt containing the order details, amount paid, and loyalty points.
10. The use case ends successfully.

Alternate Flow

- A1.** Payment declined: The system displays a payment failure message and asks the customer to retry or select another supported payment method.
- A2.** Payment fails twice: The system cancels the transaction and returns the kiosk to the ordering screen without marking the order as paid.
- A3.** Receipt printing fails after successful payment: The system displays a receipt/reprint option and retains the paid order details.

Precondition: The system is operational and the customer can access the ordering interface.

Postcondition: A successful order is stored and the customer receives the appropriate confirmation; otherwise the system preserves the order state and provides an actionable error message.